

Discrete Mathematics
MATH1031 - Durham University

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Contents

I	Module Content	1
13	Introduction to Discrete Mathematics	1
13.1	Arrangements and Permutations	1
13.2	Combinations	2
13.3	Arrangements and Combinations with Repetitions	2
14	Combinatorics	4
14.1	The Three Principles	4
14.2	Recurrence Relations	5
14.3	Generating Functions	6
15	Graph Theory	6
15.1	What is a Graph?	6
15.2	Eulerian Graphs	6
15.3	Planar Graphs	6
II	References	7
16	Module References	7
16.1	Module Professors	7
16.2	Cited Works	7

Part I

Module Content

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (1)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (2)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (3)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (4)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (5)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial. □

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded. □

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (6)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b . □

13.3 Arrangements and Combinations with Repetitions

13.3.1 Arrangements with Repetitions

Standard Problem 4: How many arrangements are there of a list of objects in which some objects are repeated?

Solution: **Theorem 13.3.1.1:** Given a list of n objects of r different types, in which objects of type i appear n_i times, the number of arrangements is

$$P(n; n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!} \quad (7)$$

Proof: Proof omitted. □

Example 13.3.1.1: How many arrangements of REARRANGE are there, and of those, how many of the arrangements have at least two R's together?

Solution: The letters to arrange are RRR AA EE G and N. Applying Theorem 13.3.1.1 with $n = 9$ and $n_1 = 3, n_2 = 2, n_3 = 2, n_4 = 1, n_5 = 1$, we get

$$\frac{9!}{3!2!2!} = 15,120 \quad (8)$$

We can then take all the cases where all the R's are separate. To do this, we arrange all the other letters, giving $\frac{6!}{2!2!}$ arrangements. We then insert the R's into the available gaps, giving 7 possible spots for 3 R's, meaning the number of arrangements are

$$\frac{9!}{3!2!2!} - \binom{7}{3} \left(\frac{6!}{2!2!} \right) = 8,820 \quad (9)$$

13.3.2 Combinations with Repetitions

Standard Problem 5: Given a set of n objects, how many combinations of size k can be chosen if objects may be selected more than once? (How many k -combinations with repetitions from n objects are there?)

Solution: To solve this problem, first consider

Standard Problem 6: How many k -combinations with repetitions from n objects are there in which each object is chosen at least once in the combination?

Solution: **Theorem 13.3.2.1:** The number of k -combinations from n different objects with repetition and each object occurring at least once is $\binom{k-1}{n-1}$ for $k \geq n$.

Proof: This question clearly has a non-zero answer only if $k \geq n$. Since order does not matter in our combinations, we represent each combination with its items in non-decreasing order. Now insert markers into the strings at the points where the numbers change. Because we have specified that each number appears at least once, there must be $(n-1)$ markers occupying $(k-1)$ possible positions in each of these lists.

□

Example 13.3.2.1: Count the 5-combinations with repetitions from $\{1, 2, 3\}$ in which each number appears at least once.

Solution: Using the method above, we get the following combinations:

$$\begin{aligned} 11123 &\leftrightarrow 111|2|3 \\ 11223 &\leftrightarrow 11|22|3 \\ 11233 &\leftrightarrow 11|2|33 \\ 12223 &\leftrightarrow 1|222|3 \\ 12233 &\leftrightarrow 1|22|33 \\ 12333 &\leftrightarrow 1|2|333 \end{aligned}$$

There are 4 possible gaps for 2 markers, so the number of ways of arranging the markers is $\binom{4}{2} = 6$.

We can adapt this solution to solve Standard Problem 5 by allowing the markers to also be placed next to one another, or outside of the item strings. Taking this into account, we get:

Theorem 13.3.2.2: The number of k -combinations from n different objects with repetition is

$$\binom{k+n-1}{n-1} = \binom{k+n-1}{k} \quad (10)$$

Proof: As explained above.

□

13.3.3 The Extended Binomial Coefficients

Proposition 13.3.3.1: The principle of upper negation states that, if n and k are positive integers, then

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k} = (-1)^k \binom{k+n-1}{n-1} \quad (11)$$

Proof:

$$\binom{-n}{k} = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!} \quad (12)$$

Factoring out a factor of -1,

$$\binom{-n}{k} = (-1)^k \frac{(n)(n+1)(n+2)\dots(n+k-1)}{k!} \quad (13)$$

Which simplifies to give the claimed result.

□

14 Combinatorics

14.1 The Three Principles

14.1.1 Mathematical Induction

Definition 14.1.1.1: Mathematical induction provides a method for establishing the truth of a sequence of statements, each of which is associated with a non-negative integer. In its simplest form, it gives efficient proofs of identities such as the binomial formula.

Theorem 14.1.1.1: The principle of mathematical induction states that: suppose that $P(n)$ is a sequence of propositions labelled by integers $n \geq n_0$. Suppose that

- a. $P(n_0)$ is true, and
- b. For any $n \geq n_0$, the truth of $P(n)$ implies the truth of $P(n+1)$.

Then, for every $n \geq n_0$, $P(n)$ is true.

Proof: $P(n_0)$ is true by assumption (a). Then by (b), $P(n_0+1)$ is also true. Again, by (b), $P(n_0+2)$ is also true, and so on. This chain of deductions eventually reaches any given $n \geq n_0$, so $P(n)$ is true for all $n \geq n_0$.

□

14.1.2 The Pidgeon-Hole Principle

Definition 14.1.2.1: The pidgeon-hole principle states that if n envelopes are distributed among m pidgeon-holes, and $n \geq m$, then at least one pidgeon-hole contains more than one envelope.

14.1.3 The Inclusion-Exclusion Principle

Proposition 14.1.3.1: We have

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \quad (14)$$

Proof: Apply $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$ repeatedly.

□

Theorem 14.1.3.1: The inclusion-exclusion formula states that, for a positive integer n and finite sets $A_1, \dots, A_n$,

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{k=1}^n \left((-1)^{k+1} \times \sum_{i_1 < i_2 < \dots < i_k} |A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}| \right) \quad (15)$$

Proof: A slightly tricky induction on n , with the base case $n = 2$ given by $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. □

Definition 14.1.3.1: A derangement of the numbers $\{1, 2, \dots, n\}$ is a permutation in which no number is in its correct position.

Proposition 14.1.3.2: The number of derangements of $1, 2, \dots, n$ is given by

$$d(n) = \sum_{k=0}^n (-1)^k \frac{n!}{k!} \quad (16)$$

Proof: We subtract from $n!$ the number of non-derangements to get

$$d(n) = n! \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (17)$$

□

In particular, the proportion of permutations that are derangements is

$$\frac{d(n)}{n!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (18)$$

This is the power series expansion of e^{-1} , so

$$\lim_{n \rightarrow \infty} \frac{d(n)}{n!} = \frac{1}{e} \quad (19)$$

14.2 Recurrence Relations

Definition 14.2.0.1: As defined by (Perepelitsa, 2020, p.1), a recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence for all integers n with $n \geq n_0$, where n_0 is a non-negative integer.

Definition 14.2.0.2: As defined by (Perepelitsa, 2020, p.1), a sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Definition 14.2.0.3: As defined by (Perepelitsa, 2020, p.1), the initial conditions for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Let $\{a_n\}$ be a sequence indexed by integers $n = 0, 1, 2, \dots$, we are given:

- The initial conditions $a_0, a_1, \dots, a_{r-1}$.
- The recurrence relation

$$a_n = \sum_{i=1}^r c_i a_{n-i} = c_1 a_{n-1} + \dots + c_r a_{n-r}, \quad (n \geq r) \quad (20)$$

where $c_1, \dots, c_r$ are specified constants.

Definition 14.2.0.4: The recurrence relation is linear because no term involves more than a single factor of the a_i , and it is homogeneous because every term involves an a_i .

Definition 14.2.0.5: We seek solutions to the recurrence relation (20) of the form $a_n = x^n$, $x \neq 0$. Substituting this into the relation, we get

$$x^n = \sum_{i=1}^r c_i x^{n-i}, \quad (n \geq r) \quad (21)$$

Dividing through by x^{n-r} , we get

$$x^r - \sum_{i=1}^r c_i x^{r-i} = 0 \quad (22)$$

This is the characteristic equation.

CONTINUE ON P.39 WHERE THE STEPS BEGIN.

14.3 Generating Functions

15 Graph Theory

15.1 What is a Graph?

15.2 Eulerian Graphs

15.3 Planar Graphs

Part II

References

16 Module References

16.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* are taught by Professor Mikhail Menshikov.

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Graph Theory is taught by Irene Ayuso-Ventura.

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16.2 Cited Works

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